

AT2 – Visualization Analysis Report

Data Handling and Visualization – DSA06

Complete R Programming Practical Answers

Below is a complete, exam-ready solution for all five questions. The R code uses the given datasets exactly and includes visualization, interpretation, comparison, and visualization-principle analysis.

Q1. Comparing Distributions – Histogram, Density and ECDF

Given Data

```
Time <- c(18, 21, 22, 24, 25, 25, 27, 28, 29, 30,  
          31, 32, 33, 35, 36, 38, 41, 43, 46, 50)
```

R Program

```
# Q1: Comparing Distributions
```

```
Time <- c(18, 21, 22, 24, 25, 25, 27, 28, 29, 30,  
          31, 32, 33, 35, 36, 38, 41, 43, 46, 50)
```

```
# Mean and median
```

```
mean(Time)
```

```
median(Time)
```

```
sd(Time)
```

```
# -----
```

```
# 1. Histogram
```

```
# -----
```

```
hist(Time,
```

```
      breaks = 6,
```

```
      col = "skyblue",
```

```
      border = "black",
```

```
main = "Histogram of Processing Times",  
xlab = "Processing Time (minutes)",  
ylab = "Frequency")
```

```
# -----
```

```
# 2. Density Plot
```

```
# -----
```

```
plot(density(Time),  
     main = "Density Plot of Processing Times",  
     xlab = "Processing Time (minutes)",  
     ylab = "Density",  
     lwd = 2)
```

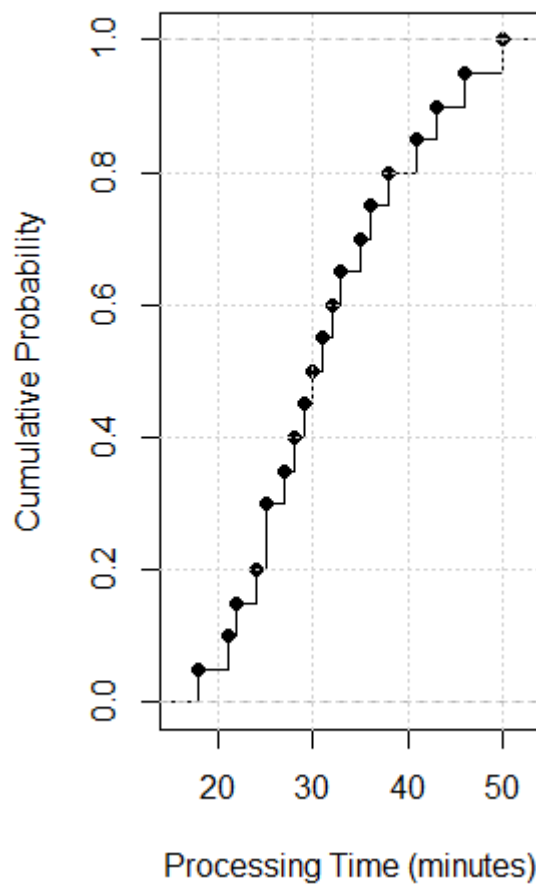
```
# -----
```

```
# 3. ECDF
```

```
# -----
```

```
plot(ecdf(Time),  
     main = "Empirical Cumulative Distribution Function",  
     xlab = "Processing Time (minutes)",  
     ylab = "Cumulative Probability",  
     verticals = TRUE,  
     do.points = TRUE)
```

Empirical Cumulative Distribution Function



grid()

Important Numerical Values

- Mean = **31.7 minutes**
- Median = **30.5 minutes**
- Standard deviation $\approx$ **8.63 minutes**
- Minimum = **18 minutes**
- Maximum = **50 minutes**
- Q1 = **25 minutes**
- Q3 = **36.5 minutes**

Interpretation

Histogram:

The histogram shows how frequently processing times occur within different time

intervals. Most observations are concentrated approximately between **24 and 38 minutes**. The distribution has a moderate right tail because some tasks take considerably longer, reaching 46–50 minutes.

Density plot:

The density plot provides a smooth representation of the distribution. It makes the central concentration around approximately **30–32 minutes** easier to identify. It is useful for observing the overall shape and possible skewness without depending heavily on histogram bin boundaries.

ECDF:

The ECDF shows the cumulative proportion of observations below or equal to each processing time. For example, approximately half of the observations are at or below the median of **30.5 minutes**. It is especially useful when the objective is to answer questions such as "What percentage of tasks are completed within 35 minutes?"

Comparison

Visualization	Center	Spread	Skewness	Cumulative Behavior
Histogram	Good	Very good	Good	Poor
Density	Very good	Good	Very good	Poor
ECDF	Good	Very good	Moderate	Excellent

Conclusion

The **histogram** is best for understanding frequencies and overall distribution. The **density plot** gives a smoother view of the distribution shape and skewness. The **ECDF** is best for understanding cumulative behavior and percentile-related questions. Using all three provides a more complete understanding of the processing-time distribution.

Q2. Q-Q Plot and Normality Analysis

Given Data

```
Response <- c(42, 45, 47, 49, 50, 51, 52, 53, 54, 55,  
              56, 57, 58, 60, 61, 63, 66, 70, 76, 85)
```

R Program

```
# Q2: Q-Q Plot and Normality Analysis
```

```
Response <- c(42, 45, 47, 49, 50, 51, 52, 53, 54, 55,  
             56, 57, 58, 60, 61, 63, 66, 70, 76, 85)
```

```
# Mean, median and standard deviation
```

```
mean(Response)
```

```
median(Response)
```

```
sd(Response)
```

```
# -----
```

```
# Histogram
```

```
# -----
```

```
hist(Response,
```

```
      breaks = 7,
```

```
      col = "lightgreen",
```

```
      border = "black",
```

```
      main = "Histogram of Response Times",
```

```
      xlab = "Response Time (seconds)",
```

```
      ylab = "Frequency")
```

```
# Add density curve
```

```
lines(density(Response),
```

```
      col = "red",
```

```
      lwd = 2)
```

```
# -----
```

```
# Density plot
```

```
# -----
```

```
plot(density(Response),
```

```
main = "Density Plot of Response Times",  
xlab = "Response Time (seconds)",  
ylab = "Density",  
lwd = 2)
```

```
# -----
```

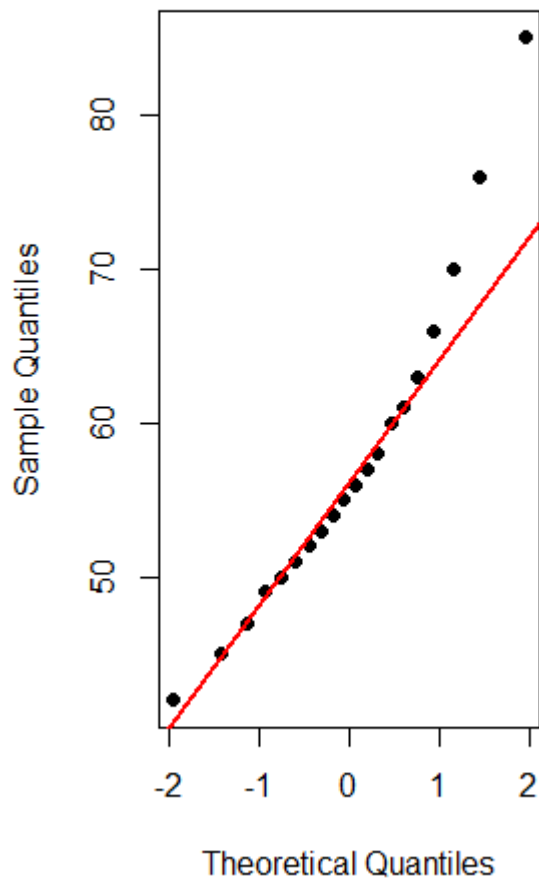
```
# Normal Q-Q Plot
```

```
# -----
```

```
qqnorm(Response,  
  main = "Normal Q-Q Plot of Response Times",  
  pch = 19)
```

```
qqline(Response,  
  col = "red",  
  lwd = 2)
```

Normal Q-Q Plot of Response Tin



Important Numerical Values

- Mean = **57.5 seconds**
- Median = **55.5 seconds**
- Standard deviation $\approx$ **10.58 seconds**
- Minimum = **42 seconds**
- Maximum = **85 seconds**

Interpretation of Histogram

The histogram shows that most response times are concentrated between approximately **49 and 66 seconds**. The observation at **85 seconds** is relatively far from the central group.

The mean (**57.5**) is greater than the median (**55.5**), indicating a **slight positive/right skew**.

Q-Q Plot Interpretation

For a normally distributed dataset, points in a Q-Q plot should approximately follow the reference straight line.

In this dataset:

- The middle observations are reasonably close to the reference line.
- The upper-tail observations, particularly **76 and 85 seconds**, deviate noticeably above the line.
- The value **85 seconds** appears to be a potential high-end outlier.
- The deviations suggest a **heavier/right tail** than would be expected under a perfectly normal distribution.

Conclusion

The data can be considered **approximately normal in the central region**, but it is not perfectly normal. The upper tail shows noticeable deviation, mainly because of the high response time of **85 seconds**. Therefore, the dataset exhibits **mild positive skewness and a possible high-end outlier**.

The Q-Q plot is more informative than the histogram for detecting departures from normality because it directly compares the observed quantiles with theoretical normal quantiles.

Q3. Visualizing Many Distributions at Once

Given Data

```
Group_A <- c(52,55,57,58,60,61,62,64,65,66,68,70,72,74,78)
```

```
Group_B <- c(45,49,51,53,55,56,58,59,60,62,63,65,67,69,71)
```

```
Group_C <- c(58,60,61,63,64,65,67,68,70,71,72,74,76,80,84)
```

R Program – Boxplot with Individual Observations

```
# Q3: Comparing Three Groups
```

```
Group_A <- c(52,55,57,58,60,61,62,64,65,66,68,70,72,74,78)
```



```
Group_B <- c(45,49,51,53,55,56,58,59,60,62,63,65,67,69,71)
```

```
Group_C <- c(58,60,61,63,64,65,67,68,70,71,72,74,76,80,84)
```

```
# Combine data
```

```
scores <- c(Group_A, Group_B, Group_C)
```

```
group <- factor(c(rep("Group A", length(Group_A)),  
                  rep("Group B", length(Group_B)),  
                  rep("Group C", length(Group_C))))
```

```
data <- data.frame(Group = group, Score = scores)
```

```
# Boxplot
```

```
boxplot(Score ~ Group,  
        data = data,  
        col = c("skyblue", "lightgreen", "orange"),  
        main = "Comparison of Examination Scores",  
        xlab = "Student Group",  
        ylab = "Examination Score")
```

```
# Overlay individual observations
```

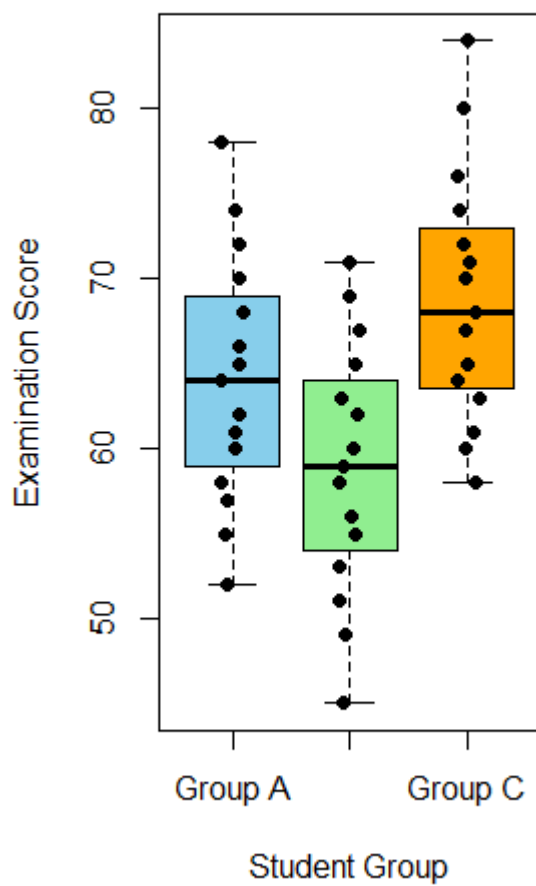
```
stripchart(Score ~ Group,  
           data = data,  
           vertical = TRUE,  
           method = "jitter",  
           add = TRUE,  
           pch = 19)
```

Optional Violin Plot Using ggplot2

```
library(ggplot2)
```

```
ggplot(data, aes(x = Group, y = Score)) +  
  geom_violin(fill = "lightblue", alpha = 0.6) +  
  geom_boxplot(width = 0.15, fill = "white") +  
  geom_jitter(width = 0.08, size = 2) +  
  labs(title = "Distribution of Examination Scores",  
        x = "Student Group",  
        y = "Examination Score") +  
  theme_minimal()
```

Comparison of Examination Scores



Numerical Comparison

Group Mean Median Q1 Q3 Minimum Maximum

A	64.13	64	59	69	52	78
B	58.87	59	54	64	45	71
C	68.87	68	63.5	73	58	84

Interpretation

Group A:

Group A has a middle score of approximately **64**. The scores range from **52 to 78**, showing moderate spread.

Group B:

Group B has the lowest center, with a median of **59**. Its scores range from **45 to 71**. Therefore, Group B generally performs lower than Groups A and C.

Group C:

Group C has the highest center, with a median of **68**. It also contains the highest score of **84**. This indicates that Group C generally performs better than the other two groups.

Outliers

The boxplot does not show any strong conventional outliers based on the $1.5 \times \text{IQR}$ rule. The extreme values such as **45 in Group B** and **84 in Group C** are relatively distant but remain within the expected distribution range.

Why One Combined Visualization Is Better

A grouped boxplot or violin plot allows all three groups to be compared using the **same scale and visual structure**.

Three separate plots would make comparison more difficult because the viewer would have to mentally compare different plots.

A combined plot immediately shows:

- Group C has the highest center.
- Group B has the lowest center.
- Group A lies between them.
- The spreads are relatively similar.
- Individual observations and possible unusual values can be identified.

Conclusion

The **grouped boxplot with jittered observations** is the most suitable choice because it simultaneously communicates **median, quartiles, spread, possible outliers, and individual observations**.

Q4. Relationships and Multivariate Visualization

Given Data

```
Advertising <- c(10,12,14,16,18,20,22,24,26,28,30,32,34,36,38)
```

```
Visits <- c(120,135,150,165,180,195,210,225,245,260,275,290,305,320,340)
```

```
Sales <- c(48,50,53,55,58,61,63,66,69,71,74,77,80,83,86)
```

R Program

```
# Q4: Multivariate Visualization
```

```
Advertising <- c(10,12,14,16,18,20,22,24,26,28,30,32,34,36,38)
```

```
Visits <- c(120,135,150,165,180,195,210,225,245,260,275,290,305,320,340)
```

```
Sales <- c(48,50,53,55,58,61,63,66,69,71,74,77,80,83,86)
```

```
data <- data.frame(Advertising, Visits, Sales)
```

```
# Scatterplot using point size for website visits
```

```
plot(Advertising, Sales,
```

```
  pch = 19,
```

```
  cex = Visits / 100,
```

```
  main = "Advertising Expenditure vs Sales",
```

```
  xlab = "Advertising Expenditure",
```

```
  ylab = "Sales")
```

```
# Add fitted trend line
```

```
model <- lm(Sales ~ Advertising, data = data)
```

```
abline(model,
```

```
  col = "red",
```

```
  lwd = 2)
```

```
# Correlation
```

```
cor(Advertising, Sales)
```

```
# Regression summary
```

```
summary(model)
```

```
ggplot2 Version
```

```
library(ggplot2)
```

```
ggplot(data,
```

```
  aes(x = Advertising,
```

```
    y = Sales,
```

```
    size = Visits)) +
```

```
geom_point(alpha = 0.7) +
```

```
geom_smooth(method = "lm",
```

```
  se = FALSE) +
```

```
labs(title = "Advertising Expenditure vs Sales",
```

```
  subtitle = "Point size represents Website Visits",
```

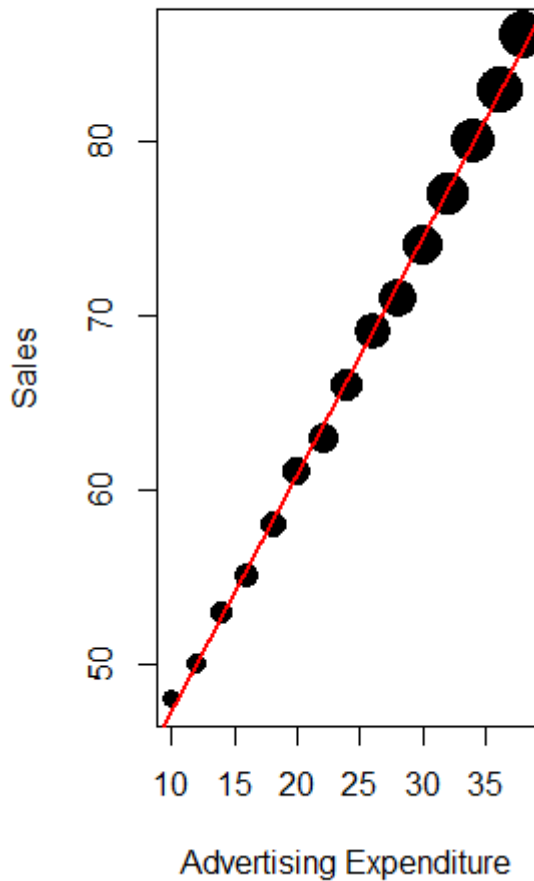
```
  x = "Advertising Expenditure",
```

```
  y = "Sales",
```

```
  size = "Website Visits") +
```

theme_minimal()

Advertising Expenditure vs Sales



Relationship Analysis

The Pearson correlation between advertising expenditure and sales is approximately:

$r = 0.9994$

This indicates an **extremely strong positive linear relationship** between advertising expenditure and sales.

The fitted regression equation is approximately:

$$Sales = 33.65 + 1.36(Advertising)$$

Therefore, an increase of one unit in advertising expenditure is associated with approximately **1.36 units of increase in sales** in this dataset.

The scatterplot points lie very close to the fitted trend line, confirming the strong linear relationship.

Role of Website Visits

Website visits increase along with advertising expenditure. Encoding website visits using **point size** provides an additional dimension of information.

This helps the viewer observe that:

- Higher advertising expenditure is associated with higher website visits.
- Higher website visits occur alongside higher sales.
- The three variables show a broadly positive pattern.

Does the Additional Variable Help?

Yes, but only if used carefully.

Using point size for website visits provides useful additional information without requiring a separate graph. However, excessive visual encodings can make a scatterplot difficult to read.

For this dataset, **point size is acceptable**, but a very large range of point sizes or several additional colors/shapes could create clutter.

Conclusion

The scatterplot with a fitted regression line is the most appropriate primary visualization because it clearly communicates the strong positive relationship between advertising expenditure and sales. Website visits can be incorporated through point size, but the additional encoding should remain simple to avoid visual clutter.

Important: The strong correlation shows association, but it does not by itself prove that advertising expenditure causes the increase in sales.

Q5. Misleading Visualization and Redesign

Given Data

```
Department <- c("CSE", "ECE", "EEE", "MECH", "CIVIL", "AIDS", "IT")
```

```
Pass_Percentage <- c(91, 93, 92, 89, 90, 95, 94)
```

Part 1 – Misleading Visualization

A truncated y-axis can exaggerate relatively small differences.

R Code

Q5: Misleading Visualization

```
Department <- c("CSE", "ECE", "EEE", "MECH", "CIVIL", "AIDS", "IT")
```

```
Pass_Percentage <- c(91, 93, 92, 89, 90, 95, 94)
```

```
data <- data.frame(Department, Pass_Percentage)
```

Misleading plot: truncated y-axis

```
barplot(Pass_Percentage,
```

```
  names.arg = Department,
```

```
  col = "red",
```

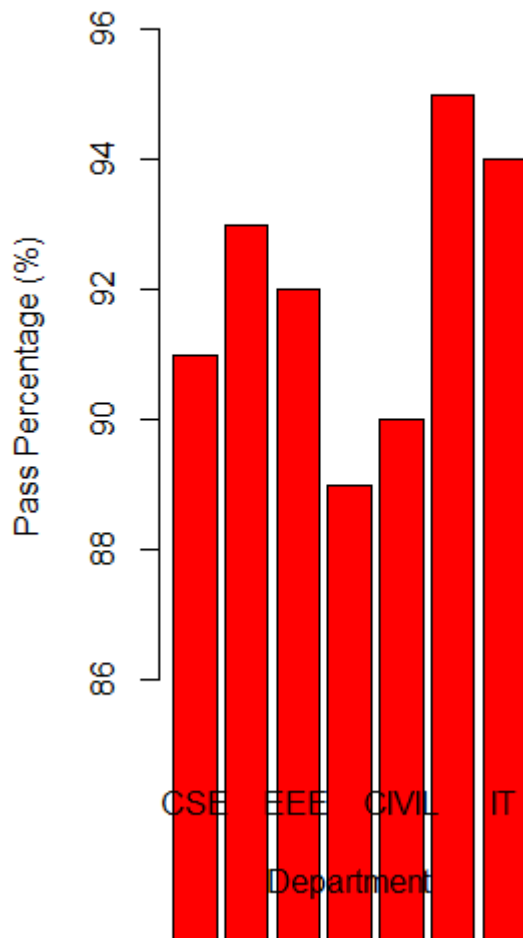
```
  ylim = c(85, 96),
```

```
  main = "Pass Percentage by Department - Truncated Axis",
```

```
  xlab = "Department",
```

```
  ylab = "Pass Percentage (%)")
```


Percentage by Department - Truncated



Why This Is Misleading

The y-axis begins at **85% instead of 0%**.

The actual difference between the lowest value (**89%**) and highest value (**95%**) is only:

$$95 - 89 = 6 \text{ percentage points}$$

However, because the graph displays only a small portion of the y-axis, the difference may visually appear much larger than it really is.

This can cause the viewer to believe that the departments have substantially different performance levels when, in fact, all departments have relatively high pass percentages.

Part 2 – Redesigned Visualization

The improved visualization should:

1. Use a baseline of **0%**.
2. Order departments according to pass percentage.
3. Display clear labels.
4. Use appropriate visual emphasis.
5. Avoid unnecessary decoration.

R Code

```
# Order departments by pass percentage
```

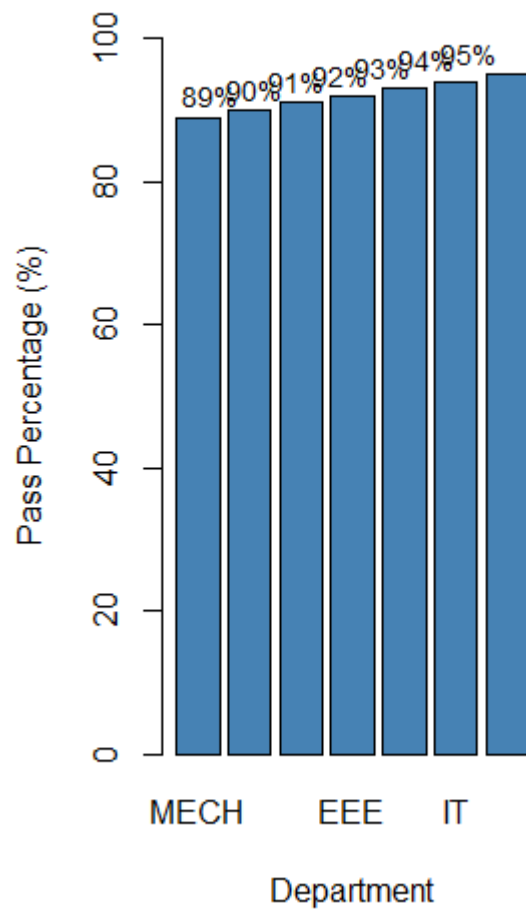
```
data_ordered <- data[order(data$Pass_Percentage), ]
```

```
barplot(data_ordered$Pass_Percentage,  
        names.arg = data_ordered$Department,  
        ylim = c(0, 100),  
        col = "steelblue",  
        border = "black",  
        main = "Pass Percentage by Department",  
        xlab = "Department",  
        ylab = "Pass Percentage (%)")
```

```
# Add percentage labels
```

```
text(x = seq_along(data_ordered$Pass_Percentage),  
     y = data_ordered$Pass_Percentage + 3,  
     labels = paste0(data_ordered$Pass_Percentage, "%"),  
     cex = 0.9)
```

Pass Percentage by Department



Ordered Data

Rank Department Pass Percentage

1	MECH	89%
2	CIVIL	90%
3	CSE	91%
4	EEE	92%
5	ECE	93%
6	IT	94%
7	AIDS	95%

Interpretation

The redesigned chart shows that all departments have **high pass percentages ranging from 89% to 95%**.

- **AIDS** has the highest pass percentage at **95%**.
- **IT** follows with **94%**.
- **ECE** has **93%**.
- **EEE** has **92%**.
- **CSE** has **91%**.
- **CIVIL** has **90%**.
- **MECH** has the lowest value at **89%**.

Although the departments differ, the overall difference is only **6 percentage points**.

Comparison of the Two Visualizations

Feature	Misleading Plot	Redesigned Plot
Y-axis	Starts at 85%	Starts at 0%
Difference appearance	Exaggerated	Proportional
Ordering	Original order	Ordered by performance
Labels	Less informative	Clear percentage labels
Interpretation	Can mislead	More honest
Accessibility	Potentially confusing	Easier to understand

Accessibility and Communication

The redesigned visualization should use:

- Clear axis labels.
- A readable font size.
- Sufficient contrast.
- Direct percentage labels.
- No unnecessary 3D effects.
- A simple, consistent color.
- A meaningful title.

Color should not be the only method used to distinguish important information because some viewers may have color-vision deficiencies.

Final Conclusion

The first visualization is potentially misleading because the truncated y-axis makes a **6-percentage-point difference appear visually dramatic**. The redesigned chart uses a zero baseline and ordered categories, allowing the viewer to understand the actual magnitude of differences more honestly.

Overall Conclusion

The five visualization tasks demonstrate that choosing the correct visualization is essential for accurate data communication.

1. **Histogram, density plot and ECDF** provide complementary views of a distribution.
2. **Q-Q plots** are particularly useful for assessing normality and detecting tail deviations.
3. **Grouped boxplots/violin plots** allow distributions to be compared efficiently using a common scale.
4. **Scatterplots with trend lines** clearly communicate relationships between numerical variables, while additional encodings should be used carefully.
5. **Appropriate axis scaling and ordering** are critical to prevent misleading interpretations.

Therefore, effective visualization is not only about producing attractive graphs; it is about presenting data in a way that is **accurate, interpretable, honest, and appropriate for the analytical question**.